SCAN:

VERILOG:

```
module scan_reg_8bit (
    input clk,
    input rstn,              // Active LOW reset
    input [7:0] d,
    input scan_in,
    input scan_en,
    output reg [7:0] q,
    output scan_out
);
integer i;
always @(posedge clk) begin
    if (!rstn)
        q <= 8'b00000000;
    else if (scan_en) begin
        q[0] <= scan_in;
        for (i = 1; i < 8; i = i + 1)
            q[i] <= q[i-1];
    end
    else
        q <= d;
end
assign scan_out = q[7];
endmodule
```

TEST BENCH:

```
module tb_scan_reg;
reg clk;
reg rstn;
reg [7:0] d;
reg scan_in, scan_en;
wire [7:0] q;
wire scan_out;
scan_reg_8bit uut (
    .clk(clk),
    .rstn(rstn),
    .d(d),
    .scan_in(scan_in),
    .scan_en(scan_en),
    .q(q),
    .scan_out(scan_out)
);
// Clock
always #5 clk = ~clk;
initial begin
```

```
    $dumpfile("scan_activity.vcd");
    $dumpvars(0, tb_scan_reg);
    clk = 0;
    rstn = 0;
    scan_en = 0;
    scan_in = 0;
    d = 8'b00000000;
    #20 rstn = 1;
    //  Functional Mode
    scan_en = 0;
    #10 d = 8'b11001100;
    #10 d = 8'b10101010;
    //  Scan Mode
    scan_en = 1;
    scan_in = 1; #10;
    scan_in = 0; #10;
    scan_in = 1; #10;
    scan_in = 1; #10;
    scan_in = 0; #10;
    scan_in = 0; #10;
    scan_in = 1; #10;
    scan_in = 0; #10;
    // Back to normal mode
    scan_en = 0;
    #20 d = 8'b11110000;
    // Reset again (extra switching)
    #20 rstn = 0;
    #10 rstn = 1;
    #30 $finish;
end
endmodule
```